

Alireza Laalmanesh

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Academic Background

University of Waterloo

Master of Mathematics (MMath) in Computational Mathematics (Co-op)

- Supervised by **Dr. David Del Rey Fernández** & **Dr. Leia Minaker**.

Waterloo, Canada

Sep. 2025 – Present

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Industrial Engineering

- Focus on Systems Analysis, Optimization, and Supply Chain Management.
- Built a strong foundation in mathematical modeling and engineering statistics.

Tehran, Iran

Sep. 2020 – Jun. 2025

Honors & Awards

i-Comp-Data Fellowship

- Supporting high-impact, interdisciplinary research projects bridging Mathematics and Environmental Planning.

Provost-funded Award

2025

Top 1% Rank

- Achieved top ranking among +150,000 candidates nationwide.

National University Entrance Exam

2020

Merit-based Scholarship

- Awarded for outstanding academic performance during undergraduate studies.

Amirkabir University of Technology

2020 – 2025

Professional Experience

University of Waterloo

Graduate Teaching Assistant

- **MATH 128:** Calculus 2 for the Sciences | **MATH 106:** Applied Linear Algebra 1.
- Facilitating tutorials and supporting students in mastering fundamental calculus and algebraic concepts.

Waterloo, Canada

Jan. 2026 – Present

University of Waterloo

Graduate Teaching Assistant

- **MATH 135:** Algebra for Honours Mathematics | **MATH 115:** Linear Algebra for Engineering.
- Assisted in grading and provided feedback on mathematical proofs and engineering-focused linear algebra.

Waterloo, Canada

Sep. 2025 – Dec. 2025

Self-employed (Private)

Mathematics & Physics Tutor

- Focused on fostering deep conceptual understanding in Math and Physics for grades 7–12.
- Tailored instructional methods to individual learning styles to enhance problem-solving skills.

Tehran, Iran

Dec. 2019 – Sep. 2025

Amirkabir University of Technology

Teaching Assistant

- Assisted in "Introduction to Probability Theory", "Optimization", and "Systems Analysis".
- Designed mathematical modeling assignments to bridge theoretical concepts with practical applications.

Tehran, Iran

Sep. 2023 – Jun. 2025

TAV Institute (Tavenergyco)

Factory Layout Design Intern

- Applied theoretical optimization techniques to address real-world supply chain and layout challenges.
- Conducted capacity analysis to identify and mitigate production bottlenecks.

Tehran, Iran

Jul. 2023 – Dec. 2023

Kanoon Farhangi Amoozesh (Galamchi)

Academic Advisor

- Guided students in academic planning, goal setting, and developing efficient study habits.

Tehran, Iran

Dec. 2019 – Dec. 2020

Recent Experiments (Repositories)

Traffic Severity Study | *PyTorch, DeepInsight, Custom Loss*

- A study on imbalanced data challenges. Experimenting with custom **Soft F-Beta Loss** functions to improve minority class representation.

CrashVision DL Exploration | *CV, Feature Engineering, AI*

- Exploring Computer Vision approaches for crash severity prediction. Practicing feature mapping utilizing the **DeepInsight** methodology.

LinguoMath TTS Experiment | *Python, Asyncio, Regex*

- An experimental pipeline for parsing scientific PDFs. Investigating ways to translate complex $\text{\LaTeX}$ formulas into natural language via Asyncio.

Android Hardware Bridge | *Low-level Protocols, ADB, Networking*

- A personal project exploring **USB/IP protocols** to repurpose rooted Android devices as headless servers over TCP/IP networks.

CodeBase Aggregator | *Python, Automation, File I/O*

- A utility script designed to aggregate repositories into datasets. Built to better understand LLM context injection and automation.

Bilingual Audio Testing | *Audio Processing, AI, Dubbing*

- Experimenting with AI voice synthesis APIs to automate the translation of educational video audio.

EagleEye Crawler | *Selenium, AsyncIO, Data Mining*

- Learning asynchronous web scraping and data collection strategies by exploring Python's AsyncIO and Selenium capabilities.

Knowledge Journey

Tools for Implementation: Python, C/C++ (Algorithm Design)

Academic Explorations: Numerical Optimization, Stochastic Modeling, Loss Function Design

AI & Simulation: PyTorch, TensorFlow, EfficientNet, CNNs, Discrete Event Simulation (DES)

Systems & Automation: Docker, Low-level Protocols, ADB, Hardware Protocols, Asyncio Automation